

KYAW SOE LWIN

Data Science & AI/ML Engineer

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PROFESSIONAL SUMMARY

Data Science & AI/ML Engineer with deep hands-on expertise building production machine learning pipelines, Retrieval-Augmented Generation (RAG) systems, and real-time computer vision engines. Proven track record in parameter-efficient fine-tuning (PEFT/LoRA), leakage-free tabular modeling (0.844 ROC-AUC), and cloud deployment across AWS and GCP (Vertex AI). Backed by a 4.0 GPA at UC San Diego, national Jack Kent Cooke Semifinalist honors, and 9 verified industry credentials in Generative AI and Deep Learning.

PROFESSIONAL EXPERIENCE

Data Science Volunteer — Data Science Alliance

San Diego, CA (Part-time) · Sep 2026 – Present

- Contributing to a public-interest data science project analyzing longitudinal unsheltered homelessness data across Downtown San Diego (2012–present) in partnership with municipal stakeholders.
- Designed and executed an end-to-end data auditing and validation pipeline, cross-referencing multi-year counts against source reports to ensure high data integrity (97.5%+ fidelity).
- Standardized schemas and built geospatial crosswalks across 380+ downtown blocks and neighborhood boundaries to enable spatial panel modeling.
- Preparing datasets for time-series decomposition, spatial hotspot analysis (Getis-Ord Gi*), and predictive forecasting models. *Tools: Python, Pandas, NumPy, GeoJSON, Asana.*

Academic Tutor — Teacher Ni Language Centre

International · Oct 2022 – Dec 2023

- Mentored cohorts of 40+ students through structured technical curricula, conducting weekly evaluations and providing individualized feedback.

TECHNICAL SKILLS

Programming & Data: Python, SQL, Java, Bash/Linux, Pandas, NumPy, Feature Engineering

Machine Learning: Scikit-Learn, XGBoost, Optuna (Bayesian Hyperparameter Tuning), SHAP Explainability, ROI Modeling

Deep Learning & Generative AI: PyTorch, TensorFlow, Keras, Hugging Face Transformers, PEFT/LoRA Fine-Tuning, Retrieval-Augmented Generation (RAG), ChromaDB, Anthropic Claude API

Cloud & MLOps: Amazon Web Services (AWS), Google Cloud Platform (Vertex AI, GKE), Docker, Kubernetes, Git/GitHub, Streamlit, CI/CD

AI-Assisted Development: Google Antigravity, Anthropic Claude (Claude Code), Agentic Coding Workflows

EDUCATION

University of California, San Diego (UCSD) — Bachelor of Science in Data Science · Class of 2028

Skyline College — Associate Studies in Data Science · GPA: 4.0/4.0 · Aug 2024 – May 2026

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming (Java), Linear Algebra, Multivariable Calculus

TECHNICAL PROJECTS

Local Expert — Offline PDF QA Engine (RAG Pipeline)

Python, Hugging Face, ChromaDB, Anthropic Claude API, Ollama, Streamlit · 2026

- Architected an end-to-end local RAG pipeline for grounded question-answering over private PDFs with page-by-page chunking, local embeddings, and Chroma vector indexing.
- Designed a swappable LLM provider interface (Claude API & local Ollama) delivering answers grounded with source file and page citations.

LoRA TinyLlama-1.1B Instruction Fine-Tuning

PyTorch, Hugging Face, PEFT/LoRA, Google Colab (T4 GPU) · 2026

- Configured Parameter-Efficient Fine-Tuning (PEFT/LoRA) on TinyLlama-1.1B-Chat, reducing trainable parameters by >95% while retaining baseline perplexity.

Telco Customer Churn Prediction & ROI Pipeline

Python, Scikit-Learn, XGBoost, Optuna, SHAP, Streamlit · 2026

- Built a leakage-free ML pipeline on 7,043 records with stratified 70/15/15 splits; 30-trial Optuna search achieved 0.844 ROC-AUC / 0.671 PR-AUC.
- Deployed a multi-tab Streamlit dashboard with SHAP explainability and cost-sensitive ROI modeling (\$20 cost / \$200 LTV).

Priceout Collective — Affordability Policy Simulator

Python, Pandas, XGBoost, Scikit-Learn, JavaScript · Building for Good Hackathon · 2026

- Modeled 1.17M household records across 4 policy dimensions, precomputing 945 scenario combinations for zero-latency interactive simulation.
- Diagnosed and eliminated target leakage in an XGBoost model, reaching 97.3% accuracy / 0.998 AUC, validated against ALICE and HUD CHAS housing benchmarks.

Real-Time Facial Emotion Detection Engine

Python, OpenCV, TensorFlow, Keras, mini-XCEPTION (FER-2013) · 2026

- Trained a lightweight mini-XCEPTION CNN with depthwise separable convolutions on FER-2013.
- Engineered a low-latency, multi-frame OpenCV inference pipeline rendering live probability distributions across 7 emotion classes.

CERTIFICATIONS & SPECIALIZED TRAINING

DeepLearning.AI, Stanford Online & AWS (2025–2026): Generative AI with Large Language Models · Deep Learning Specialization · Machine Learning Specialization · Mathematics for Machine Learning and Data Science

Google Cloud, IBM & University Programs (2025–2026): Google Cloud Skills Boost Portfolio (Vertex AI, LLM Prompting, Model Tuning) · Frontier Tech Leaders Programme (UNDP) · Applied Python & Software Development (Harvard CS50P) · Agile Scrum (IBM) · Python for Everybody (Univ. of Michigan)

HONORS & LEADERSHIP

- **Jack Kent Cooke Undergraduate Transfer Scholarship** — National Semifinalist (2026)
- **Sterling Redman Scholarship & F.L. Griffin Scholarship** — Skyline College Recipient (2025–2026)
- **Teacher Ni Language Centre** — Academic Tutor, mentored 40+ students (2022–2023)